Lab 7: Fuzzing for Input Validation Bugs

Name: Sanjana S Aithal **SRN:** PES1UG24CS421

Application: course_registration_parser.py (Lab 6 course registration problem) **Tool:** Hypothesis + pytest

Repository: https://github.com/SanjanaSAithal/SE-Labs-PES1UG24CS421/tree/main/Lab7

The screenshots are images generated from the recorded terminal output (also committed as .txt files), not captures of a terminal window.

Bugs found by fuzzing (minimal inputs after Hypothesis shrinking)

Function	Failing input	Exception	Fix
format_course_code	None	AttributeError	Reject non-strings
format_course_code	''	ValueError (unpack)	Require exactly two words; letters-only dept, digits-only number
total_credits	None	AttributeError	Reject non-strings
total_credits	'' (also '4,,3')	ValueError (int(''))	Each field must be 1-2 ASCII digits; caps int() input size
parse_student_record	None	AttributeError	Reject non-strings
parse_student_record	'' and '|'	IndexError	Require exactly three non-empty fields

Invalid input now returns None instead of raising, and main() prints an 'Invalid ...' message. The fixed module also passed 20,000 generated examples per test.

SS1: Execution of the original application (valid input, then three malformed inputs that crash it)

SS1 – python course_registration_parser.py (original)

```
$ python course_registration_parser.py
==== Running format_course_code ====
Enter a course code as DEPT NUMBER (e.g. cs 301): cs 301
Formatted course code: CS301

==== Running total_credits ====
Enter course credits separated by commas (e.g. 4,3,4): 4,3,4
Total credits: 11

==== Running parse_student_record ====
Enter a student record as SRN|name|status (e.g. pes1ug23cs001|Asha|regular): pes1ug23cs001|Asha|regular
Student record: {'srn': 'PES1UG23CS001', 'name': 'Asha', 'status': 'regular'}

$ python course_registration_parser.py
==== Running format_course_code ====
Enter a course code as DEPT NUMBER (e.g. cs 301): cs301
Traceback (most recent call last):
  File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/buggy/co
  urse_registration_parser.py", line 40, in <module>
    main()
    ~~~~^^
  File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/buggy/co
  urse_registration_parser.py", line 29, in main
    print("Formatted course code:", format_course_code(value))
                                    ~~~~~~~~~~~~~~~~~~^^^^^^^
  File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/buggy/co
  urse_registration_parser.py", line 6, in format_course_code
    dept, number = code.split()
    ^^^^^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)

$ python course_registration_parser.py
==== Running format_course_code ====
Enter a course code as DEPT NUMBER (e.g. cs 301): cs 301
Formatted course code: CS301

==== Running total_credits ====
Enter course credits separated by commas (e.g. 4,3,4): 4,,3
Traceback (most recent call last):
  File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/buggy/co
  urse_registration_parser.py", line 40, in <module>
    main()
    ~~~~^^
  File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/buggy/co
  urse_registration_parser.py", line 33, in main
    print("Total credits:", total_credits(value))
                            ~~~~~~~~~~~~~^^^^^^^
  File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/buggy/co
  urse_registration_parser.py", line 14, in total_credits
    return sum(int(credit) for credit in text.split(','))
  File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/buggy/co
  urse_registration_parser.py", line 14, in <genexpr>
    return sum(int(credit) for credit in text.split(','))
               ~~~^^^^^^^^
ValueError: invalid literal for int() with base 10: ''

$ python course_registration_parser.py
==== Running format_course_code ====
Enter a course code as DEPT NUMBER (e.g. cs 301): cs 301
Formatted course code: CS301

==== Running total_credits ====
Enter course credits separated by commas (e.g. 4,3,4): 4,3,4
Total credits: 11

==== Running parse_student_record ====
Enter a student record as SRN|name|status (e.g. pes1ug23cs001|Asha|regular): pes1ug23cs001|Asha
Traceback (most recent call last):
  File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/buggy/co
  urse_registration_parser.py", line 40, in <module>
    main()
    ~~~~^^
  File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/buggy/co
  urse_registration_parser.py", line 37, in main
    print("Student record:", parse_student_record(value))
```

```
                           ~~~~~~~~~~~~~~~~~~~~^^^^^^^
  File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/buggy/co
  urse_registration_parser.py", line 24, in parse_student_record
    "status": parts[2].strip().lower()}
              ~~~~~^^^
IndexError: list index out of range
```

SS2: Failing test cases identified during fuzz testing (original application)

SS2 – pytest test_course_registration_parser.py (original)

```
============================= test session starts ==============================
platform darwin -- Python 3.15.0, pytest-9.2.0, pluggy-1.7.0
rootdir: Lab7
plugins: hypothesis-6.170.15
collected 3 items

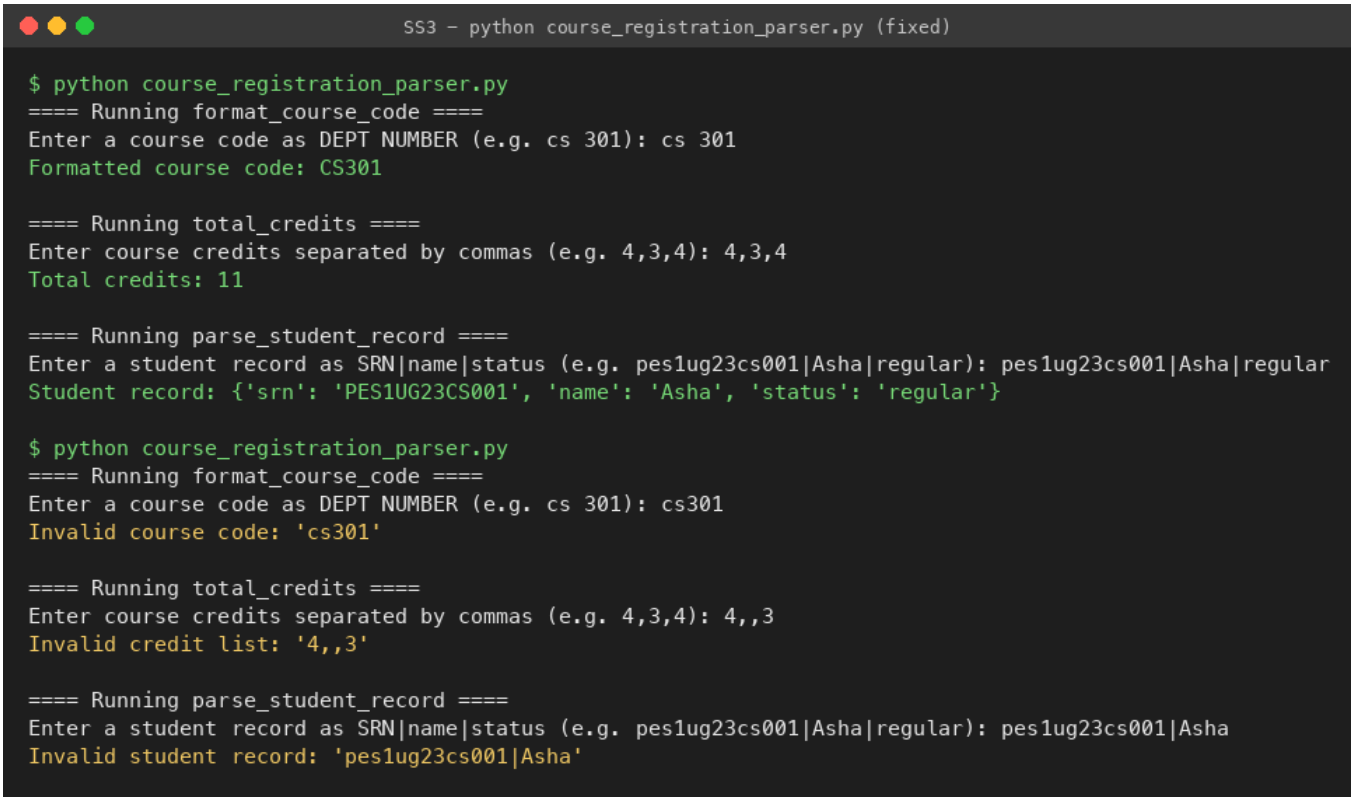
test_course_registration_parser.py FFF                                   [100%]

=================================== FAILURES ===================================
_____________________ test_format_course_code_no_crash _____________________
  + Exception Group Traceback (most recent call last):
  |   File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/my_r
  epo/Lab7/test_course_registration_parser.py", line 5, in test_format_course_code_no_crash
  |     def test_format_course_code_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/my
  _repo/Lab7/test_course_registration_parser.py", line 7, in test_format_course_code_no_crash
    |     result = format_course_code(s)
    |   File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/my
  _repo/Lab7/course_registration_parser.py", line 6, in format_course_code
    |     dept, number = code.split()
    |     ^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 0)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/my
  _repo/Lab7/test_course_registration_parser.py", line 9, in test_format_course_code_no_crash
    |     assert False, f"format_course_code crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: format_course_code crashed on '' with: ValueError: not enough values to unpack (expected 2, got 0)
    | assert False
    | Failing test case: test_format_course_code_no_crash(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         /private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/my
  _repo/Lab7/test_course_registration_parser.py:8
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/my
  _repo/Lab7/test_course_registration_parser.py", line 7, in test_format_course_code_no_crash
    |     result = format_course_code(s)
    |   File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/my
  _repo/Lab7/course_registration_parser.py", line 6, in format_course_code
    |     dept, number = code.split()
    |                    ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/my
  _repo/Lab7/test_course_registration_parser.py", line 9, in test_format_course_code_no_crash
    |     assert False, f"format_course_code crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: format_course_code crashed on None with: AttributeError: 'NoneType' object has no attribute 'split
  '
    | assert False
    | Failing test case: test_format_course_code_no_crash(
    |     s=None,
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         /private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/my
  _repo/Lab7/test_course_registration_parser.py:8
    +------------------------------------
_____________________ test_total_credits_no_crash ______________________
  + Exception Group Traceback (most recent call last):
  |   File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/my_r
  epo/Lab7/test_course_registration_parser.py", line 14, in test_total_credits_no_crash
  |     def test_total_credits_no_crash(s):
```

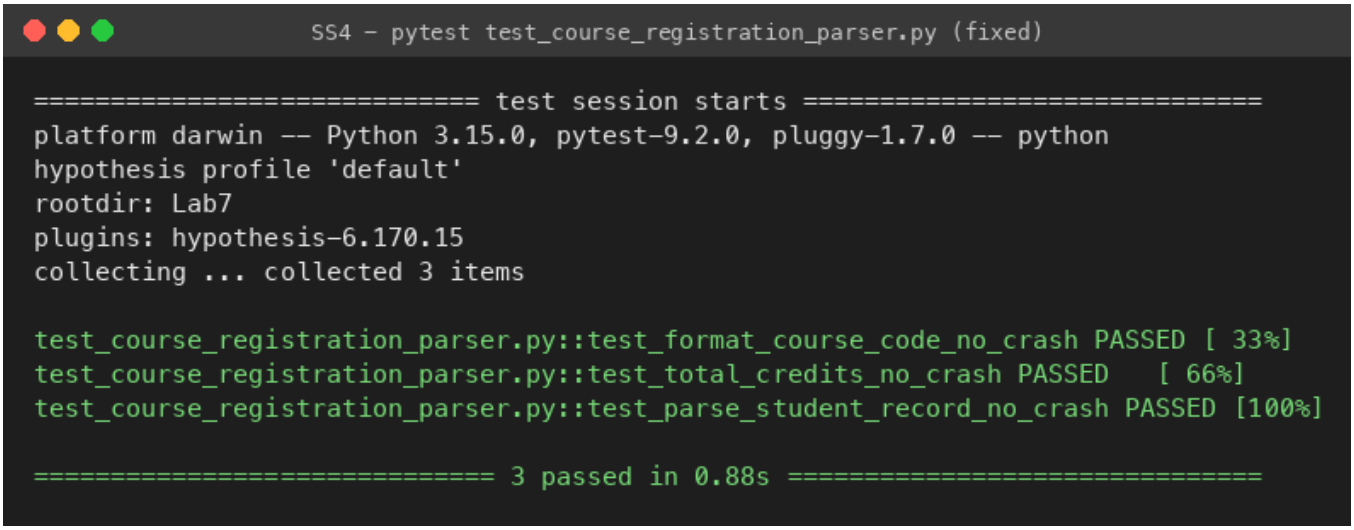
```
  |                     ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/my
  _repo/Lab7/test_course_registration_parser.py", line 16, in test_total_credits_no_crash
    |     result = total_credits(s)
    |   File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/my
  _repo/Lab7/course_registration_parser.py", line 14, in total_credits
    |     return sum(int(credit) for credit in text.split(','))
    |   File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/my
  _repo/Lab7/course_registration_parser.py", line 14, in <genexpr>
    |     return sum(int(credit) for credit in text.split(','))
    |                ~~~^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/my
  _repo/Lab7/test_course_registration_parser.py", line 18, in test_total_credits_no_crash
    |     assert False, f"total_credits crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: total_credits crashed on '' with: ValueError: invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_total_credits_no_crash(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         /private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/my
  _repo/Lab7/test_course_registration_parser.py:17
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/my
  _repo/Lab7/test_course_registration_parser.py", line 16, in test_total_credits_no_crash
    |     result = total_credits(s)
    |   File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/my
  _repo/Lab7/course_registration_parser.py", line 14, in total_credits
    |     return sum(int(credit) for credit in text.split(','))
    |                                          ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/my
  _repo/Lab7/test_course_registration_parser.py", line 18, in test_total_credits_no_crash
    |     assert False, f"total_credits crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: total_credits crashed on None with: AttributeError: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_total_credits_no_crash(
    |     s=None,
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         /private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/my
  _repo/Lab7/test_course_registration_parser.py:17
    +------------------------------------
_____________________ test_parse_student_record_no_crash _____________________
  + Exception Group Traceback (most recent call last):
  |   File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/my_r
  epo/Lab7/test_course_registration_parser.py", line 23, in test_parse_student_record_no_crash
  |     def test_parse_student_record_no_crash(s):
  |                     ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/my
  _repo/Lab7/test_course_registration_parser.py", line 25, in test_parse_student_record_no_crash
    |     result = parse_student_record(s)
    |   File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/my
  _repo/Lab7/course_registration_parser.py", line 24, in parse_student_record
    |     "status": parts[2].strip().lower()}
    |                ~~~~~^^^
    | IndexError: list index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/my
  _repo/Lab7/test_course_registration_parser.py", line 27, in test_parse_student_record_no_crash
    |     assert False, f"parse_student_record crashed on {s!r} with: {type(e).__name__}: {e}"
```

```
    | AssertionError: parse_student_record crashed on '|' with: IndexError: list index out of range
    | assert False
    | Failing test case: test_parse_student_record_no_crash(
    |     s='|',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/my
  _repo/Lab7/test_course_registration_parser.py", line 25, in test_parse_student_record_no_crash
    |     result = parse_student_record(s)
    |   File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/my
  _repo/Lab7/course_registration_parser.py", line 23, in parse_student_record
    |     "name": parts[1].strip(),
    |              ~~~~~^^^
    | IndexError: list index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/my
  _repo/Lab7/test_course_registration_parser.py", line 27, in test_parse_student_record_no_crash
    |     assert False, f"parse_student_record crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: parse_student_record crashed on '' with: IndexError: list index out of range
    | assert False
    | Failing test case: test_parse_student_record_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/my
  _repo/Lab7/test_course_registration_parser.py", line 25, in test_parse_student_record_no_crash
    |     result = parse_student_record(s)
    |   File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/my
  _repo/Lab7/course_registration_parser.py", line 21, in parse_student_record
    |     parts = line.split('|')
    |             ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/private/tmp/claude-501/-Users-sanjana-simple-shell-cli/53c2bfa0-9950-4cbb-983b-b05c986f30f8/scratchpad/my
  _repo/Lab7/test_course_registration_parser.py", line 27, in test_parse_student_record_no_crash
    |     assert False, f"parse_student_record crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: parse_student_record crashed on None with: AttributeError: 'NoneType' object has no attribute 'spl
  it'
    | assert False
    | Failing test case: test_parse_student_record_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
========================== short test summary info ===========================
FAILED test_course_registration_parser.py::test_format_course_code_no_crash
FAILED test_course_registration_parser.py::test_total_credits_no_crash - Exce...
FAILED test_course_registration_parser.py::test_parse_student_record_no_crash
============================== 3 failed in 1.39s ==============================
```

SS3: Execution of the corrected application (same valid and malformed inputs)



SS4: Final test execution showing successful results (corrected application)



course_registration_parser.py (fixed)

```
MAX_CREDIT_DIGITS = 2

def _is_ascii_digits(text):
    # str.isdigit() also accepts non-ASCII digits (superscripts, Arabic-Indic digits)
    # that int() rejects or that are not valid course numbers
    return text.isascii() and text.isdigit()

def format_course_code(code):
    """
    Formats 'cs 301' as 'CS301'.
    Returns None unless code has exactly two words: a letters-only
    department and a digits-only number.
    """
    if not isinstance(code, str):
        return None
    words = code.split()
    if len(words) != 2:
        return None
    dept, number = words
    if not (dept.isascii() and dept.isalpha() and _is_ascii_digits(number)):
        return None
    return dept.upper() + number

def total_credits(text):
    """
    Adds up comma-separated course credits.
    Returns None unless every value is a whole number of at most
    MAX_CREDIT_DIGITS digits.
    """
    if not isinstance(text, str):
        return None
    credits = [credit.strip() for credit in text.split(',')]
    for credit in credits:
        # The length cap also keeps int() away from huge digit strings, which
        # Python refuses to convert (ValueError) and which are slow to parse.
        if not _is_ascii_digits(credit) or len(credit) > MAX_CREDIT_DIGITS:
            return None
    return sum(int(credit) for credit in credits)

def parse_student_record(line):
    """
    Parses 'SRN|name|status' into a dictionary.
    Returns None unless there are exactly three non-empty fields.
    """
    if not isinstance(line, str):
        return None
    parts = [part.strip() for part in line.split('|')]
    if len(parts) != 3 or not all(parts):
        return None
    return {"srn": parts[0].upper(),
            "name": parts[1],
            "status": parts[2].lower()}

def main():
    print("==== Running format_course_code ====")
    value = input("Enter a course code as DEPT NUMBER (e.g. cs 301): ")
    result = format_course_code(value)
    if result is None:
        print("Invalid course code:", repr(value))
    else:
        print("Formatted course code:", result)

    print("\n==== Running total_credits ====")
    value = input("Enter course credits separated by commas (e.g. 4,3,4): ")
    result = total_credits(value)
    if result is None:
        print("Invalid credit list:", repr(value))
    else:
        print("Total credits:", result)

    print("\n==== Running parse_student_record ====")
    value = input("Enter a student record as SRN|name|status (e.g. pes1ug23cs001|Asha|regular): ")
    result = parse_student_record(value)
    if result is None:
        print("Invalid student record:", repr(value))
    else:
        print("Student record:", result)
```

```
if __name__ == "__main__":
    main()
```

test_course_registration_parser.py (completed)

```
from hypothesis import given, strategies as st
from course_registration_parser import format_course_code, total_credits, parse_student_record

@given(st.text() | st.none())
def test_format_course_code_no_crash(s):
    try:
        result = format_course_code(s)
    except Exception as e:
        assert False, f"format_course_code crashed on {s!r} with: {type(e).__name__}: {e}"
    # Valid input gives the formatted code; anything else gives None.
    assert result is None or isinstance(result, str)

@given(st.text() | st.none())
def test_total_credits_no_crash(s):
    try:
        result = total_credits(s)
    except Exception as e:
        assert False, f"total_credits crashed on {s!r} with: {type(e).__name__}: {e}"
    # Valid input gives a non-negative whole number; anything else gives None.
    assert result is None or (isinstance(result, int) and result >= 0)

@given(st.text() | st.none())
def test_parse_student_record_no_crash(s):
    try:
        result = parse_student_record(s)
    except Exception as e:
        assert False, f"parse_student_record crashed on {s!r} with: {type(e).__name__}: {e}"
    # Valid input gives a dict with exactly the three fields; anything else gives None.
    assert result is None or set(result) == {"srn", "name", "status"}
```